

Topic 4. NATURAL DISASTERS AND DISASTER RESPONSE.

Diseased nature oftentimes breaks forth In strange eruptions: oft the teeming earth

Is with a kind of colic pinch'd and vex'd By the imprisoning of unruly wind

*Within her womb; which, for enlargement striving,
Shakes the old beldam earth, and topples down
Steeples, and moss-grown towers.*

(William Shakespeare, 1598, King Henry IV)



All across the world, ...increasingly dangerous weather patterns and devastating storms are abruptly putting an end to the long-running debate over whether or not climate change is real. Not only is it real, it's here, and its effects are giving rise to a frighteningly new global phenomenon: the man-made natural disaster.

(Barack Obama, 2008. "An American Story: The Speeches of Barack Obama: a Primer")

SPEAKING

Brainstorm natural disasters. Write them down. In groups decide what the causes are and what a person should or should not do in the event of such natural disasters. When you are ready, share with the rest of class. Which measures are the most effective?

Notes

LISTENING COMPREHENSION

Listen to the following episode (16 minutes) and answer the questions that follow:

<https://www.bbc.co.uk/sounds/play/w3ct1j59>

BBC Business Daily September 27, 2021

Climate weighs on German elections

1. How often do such devastating floods happen in this area?

2. Describe the damage of Martina's shop. The following words might help you: ***piles of rubble, sheet rock, light sockets, till, wires.***

3. How much money was allocated by the government to help businesses recover?

4. Where else did help come from?

5. According to the podcast, what caused human loss?

6. What are the challenges rebuilding the infrastructure?

7. What are the development goals of OECD?

8. What damage can result from a major Paris flood?

9.What is the port revitalization project in Offenbach about?

10. Beside floods, what other natural disasters should be mitigated in Germany?

11.What is the example of the area where the new types of infrastructure have been implemented?

TOPIC RELATED VOCABULARY

Read the following sentences and make sure you remember the words in bold and are able to explain their meaning.

1. The process of **mitigation**: plans should be made and actions taken to **eliminate** or **reduce the threat** of future death and destruction.

2.Earth experiences **earthquakes, volcanic eruptions**, landslides, floods, fires, etc. every day. These events are common, but their **magnitudes** vary significantly. The correlation between the magnitude and the **frequency** is that frequent occurrences are low in magnitude. Another way of looking at it is to match a given magnitude with its **return period, or occurrence interval**: in general, the larger the event, the longer the return period.

3. **Natural hazards** are **inevitable** but **natural disasters** are not.

4.We need to learn how to build **disaster-resistant communities**.

DID YOU KNOW?

Earthquakes are the biggest killers followed by tropical storms and floods. During the past 500 years, global earthquakes, which occurred in Italy, Japan, China, Haiti, Indonesia, et al. killed about 5 million people. The deadliest earthquakes are likely to happen in mega-cities.

(Patrick L. Abbott *Natural Disasters*)

WATCH THE FOLLOWING VIDEO

Making buildings earthquake proof

<https://www.youtube.com/watch?v=c4fKBGslIZI> (4 minutes) 2016

Pre-listening question

What do you already know about making buildings earthquake-proof or earthquake resistant? (i.e. materials? properties? physical forces? building techniques? examples?)

Watch the video and take notes regarding:

- a) aim of earthquake-resistant buildings
- b) techniques/concepts/materials
- c) examples

Useful vocabulary:

NOUNS	VERBS	ADJECTIVES	collocations
intensity/magnitude of an earthquake	sway	earthquake-prone	burst of energy
reinforced concrete	shift	earthquake-resistant	to cause stress
base isolation	collapse	earthquake-proof	to suffer damage
foundation	deflect	seismic	to be retrofitted with
rubber	channel		to be buried
steel	dissipate		to detach ... from
shock-absorber	counteract		to keep/ prevent/ protect ... from VERB-ing
tuned-mass damper			to act like
shockwaves			to cause stress
shield			to suffer damage
core			to be retrofitted with

READING

South-west Iceland is shaking – and may be about to erupt (March 4, 2021)

More than 17,000 earthquakes have been recorded in the south-west of Iceland, in the Reykjanes Peninsula, during the past week. People living in the area have been advised to be extra careful due to dangers of landslides and rockfall. Many of the larger earthquakes have even been felt in Iceland's capital city, Reykjavik (where over half of the population lives), which lies only 27km away.

This has led to heightened concerns about the effects of even larger earthquakes and also of a possible eruption from the Krýsuvík volcanic system in the area.

South-west Iceland has a *track record of centuries of calm*, which we know can be broken by turbulent periods of intense earthquake activity accompanied by volcanic eruptions. It looks like we are entering the next turbulent period.

The most recent *earthquake swarm* is in fact the latest in a period of significantly heightened seismic activity that started over a year ago. The shaking of the Earth is the most obvious manifestation of the release of huge amounts of energy. But magma has also been quietly accumulating nearer to the surface – and when this happens there is increased likelihood of the surface breaking and the volcanoes erupting.

On March 3, concern rose sharply as a type of earthquake activity characteristic of the movement of magma was detected, indicating that an eruption might be imminent. The Civil Defence and other authorities have held press conferences, closed roads and heightened visual surveillance of the area above the potential eruption site. Of course, magma may move in the crust and then stop, but it is always wisest to plan for an eruption and then to *scale back* if nothing happens.

Huge uncertainties

The problem is that the last time south-west of Iceland experienced such a turbulent period of earthquakes and volcanic eruptions was in the 1300s – when there was no equipment to monitor seismic activity. There were also far fewer people around, which means that we don't really know what signals there were before eruptions occurred. So there are huge uncertainties.

However, Iceland has a world-leading network for monitoring seismic and volcanic unrest, and an excellent track record of anticipating eruptions and of maintaining the safety of its population. So if an eruption did happen, chances are all will be well.

An eruption in this area will be nothing like the highly disruptive explosive eruption of Eyjafjallajökull in 2010, or the much larger explosive but much less disruptive explosive eruption of Grímsvötn in 2011. Eruptions in south-west Iceland are of a fluid rock type called basalt. This results in slow-moving streams of lava fed from gently exploding craters and cones.



Scenic view of Reykjavik. [Boyloso/Shutterstock](#)

In Iceland these are warmly called “tourist eruptions” as they are relatively safe and predictable, and offer the opportunity for many hundreds of people to witness a magical natural spectacle – the creation of new land. In the past, tourists have flocked to Iceland to witness such eruptions, but at present there is a five-day quarantine period for tourists entering Iceland due to the pandemic.

At the current area of unrest, there are no nearby habitations – it is reassuringly remote. Lava streams flowing away from the area are very unlikely to damage any property on predicted pathways, but if the lava makes its way to the sea, it will cut off a few roads.

International impact?

The biggest concern internationally about a volcanic eruption in Iceland is disruption to air travel. Not only can winds carry ash clouds swiftly towards western Europe (as we saw with the Eyjafjallajökull eruption in 2010), but ash clouds can get high into the atmosphere and into the stratosphere where commercial airlines travel across the busy Atlantic flight corridors.

But the volcanoes in south-west Iceland tend not to produce much ash, and so the risk to disruption of international air travel is considered very small. Should an eruption start, flights would be halted automatically at the Keflavík international airport, which is only 22km away, until a fuller evaluation has been carried out.

The wind direction has a major effect here, and given that the prevailing wind is from a westerly direction and Keflavík sits on the western side of this south-west peninsular, winds would be expected to carry any ash away from Keflavík. The crucial effect of wind direction was highlighted beautifully in 2010 when Keflavík airport remained open while airports across Western Europe were closed for weeks.



Volcanic landscape of south-west Iceland. Dave McGarvie, Author provided

While current COVID-19 restrictions on entering Iceland will prevent hordes of tourists from travelling to see a potential eruption, there will be plenty of Icelanders travelling to observe it. They have a quirky saying in Iceland “whilst in most countries people usually run away from volcanic eruptions, in Iceland we usually run towards them”.

Dave McGarvie, Lancaster University, <https://theconversation.com/south-west-iceland-is-shaking-and-may-be-about-to-erupt-156510>, retrieved on January 11, 2022

Task: Follow the link to volcano eruptions in 2010 Eyjafjallajökull in 2010 and 2011 Grímsvötn in 2011.

How were they different from the volcano eruption in 2021? Note down 5 interesting points for each of the eruptions: 2010, 2011, 2021.

<https://www.cbc.ca/radio/quirks/bobs-blog-iceland-volcanoes-grindavik-1.7031479>

WRITING: Test Reports

Shaking Table Model Test and Seismic Performance Analysis of a High-Rise RC Shear Wall Structure

A building developed by Wuhan Shimao Group in Wuhan, China, is a high-rise 56-story (AmE story, BrE storey) 179-meter (high) residence near the Yangtze River. To investigate its seismic performance, a shaking table test with a 1/30 scale model is carried out in Structural Laboratory in Wuhan University of Technology. The dynamic characteristics and the responses of the model subject to different seismic intensities are investigated via analyzing of shaking table test data and the observed cracking pattern of the scaled model. The shaking table model test, numerical analysis, and damage identification prove that this building is well designed and can be safely put into use. Suggestions and measures to improve the seismic performance of structures are also presented.

LANGUAGE DEVICES: Concise Technical Writing. Hyphenated Adjectives.

Relative clauses/sentences/phrases become hyphenated adjectives. Compare the abstract above with the following text:

A building developed by Wuhan Shimao Group in Wuhan, China, is a **high-rise residence with 56 stories near the Yangtze River. The building has the height of 179 m.** To investigate its seismic performance, a shaking table test with a 1/30 scale model is carried out in Structural Laboratory in Wuhan University of Technology...

Exercise Come up with more concise versions of these texts using hyphenated adjectives:

1. The test was conducted using a random sample of 500 individuals who had recently purchased our product. The participants, who were recruited through an online survey platform, were asked to complete a satisfaction survey consisting of 20 questions related to the product's performance, usability, and overall experience.

2. The load test was conducted on a steel truss bridge spanning a river. The bridge, which was designed to support vehicular traffic, was subjected to a series of static and dynamic loads to evaluate its structural integrity and its capacity to carry loads. The test aimed to identify any structural deficiencies that could affect the bridge's performance under different types of loads.

3. The shake table test was performed on a 5-story building, which was retrofitted with reinforced concrete, designed to resist seismic forces. The building, which was constructed using a combination of cast-in-place and precast concrete elements, was subjected to a series of simulated earthquake ground motions with increasing intensity. The test aimed to evaluate the building's seismic performance and damage resistance. The data collected during the test, which included acceleration,

displacement, and strain measurements, was analyzed to assess the building's behavior.

4. The 3D printed house underwent a series of tests to evaluate its structural integrity and performance. The house, which was printed using a mixture of concrete and recycled materials, was subjected to tests, including water resistance, thermal insulation, and sound insulation tests. The tests aimed to assess the house's durability and its liability to provide a comfortable living environment. The data collected during the tests, which included temperature, sound and moisture measurements, was analyzed to evaluate the house's performance.

Watch the video "The Surprising Secret of Synchronization" (20 min)

<https://www.youtube.com/watch?v=t-VPRCtiUg>

Retrieved on October 25, 2021

In the video you can find the description of the test on the Millennium Bridge – 17:00-17:30.

TASK. Write a test report based on what you learned from the video and using the following prompts:

Test objective

Identify the critical point where the number of pedestrians causes bridge to suddenly increase movement

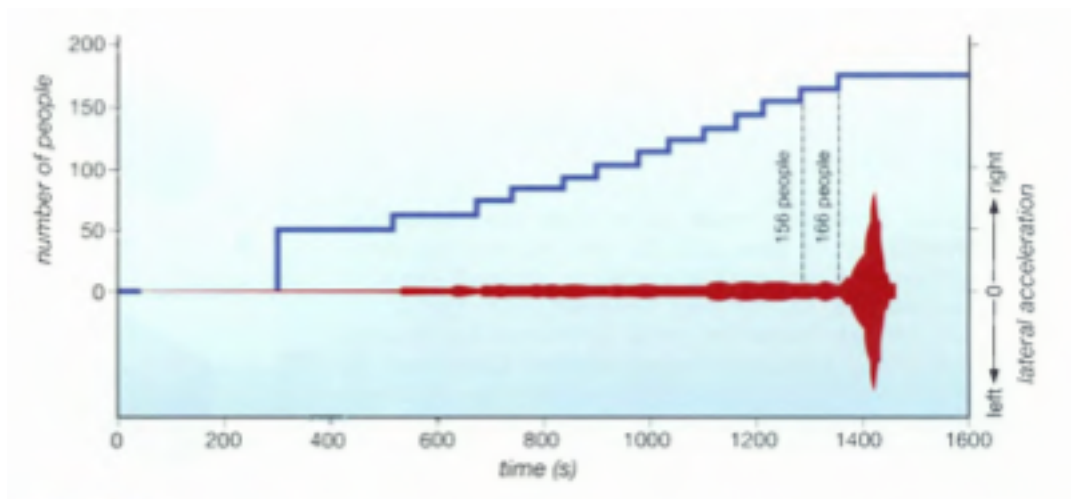
Procedure

Controlled number of pedestrians was allowed onto the bridge
increments of 2-10 starting with 50 pedestrians, walked across the bridge for 20 min
motion sensors on bridge + video cameras

Results

Critical number is 166 pedestrians

The following graph can help you as well



Please remember that a test report should have the following structure:

Introduction (Background), Test objective, Experimental setup, Procedure, Results, Conclusion.

Sometimes reports include Analysis, Further Testing, Appendix.



If you follow this structure, you will naturally write a text of about 200-300 words long.

GRAMMAR: CONDITIONALS

In the listening comprehension task on floods in Germany the following conditional sentence was used:

If I'd known the evening before, I would have got the car and brought everything home.

Which type of the conditional is it? Do you think it is this type of the conditional that is often used in technical English?

You will find the answers to these questions here

https://www.youtube.com/watch?v=huFzV5_bSn8 (2019, 8 min 30 sec).

Cause and Effect

Cause ⇒ Result

Carbon dioxide produced from burning fossil fuels ...

VERB		+ VERB
causes	the world	to warm up.
makes	the world	warm up.

The excessive use of fertilizers and pesticides can ...

VERB	+ NOUN
cause	
result in	disastrous ecological changes.
bring about	
produce	
lead to	
give rise to	

An action, idea or problem **LEADS TO** an unpleasant situation, if it begins a process which ends in that situation but may not directly cause it.

Damage to the ozone layer is **leading to** global warming.

Result ⇐ Cause

The increase in devastating storms ...

could be caused by

global warming

Smog ...

results from

a combination of smoke and fog in the air.

🔗 Prepositions after **result**: of (as a result of), in (to result in), from (to result from).

Prevention / Protection

Proper forest management

VERB

prevents

+ NOUN

deforestation.

Proper forest management

VERB

prevents
keeps
protects

FROM + NOUN/GERUND

forests **from** being destroyed.
forests **from** destruction.

protects

forests from destruction.

Exercise 1 Complete the following sentences.

1. Using inorganic fertilizers will _____ ground water pollution.
2. An increase in temperature _____ water to rise to the top of the collector.
3. Badly fitting windows _____ in cold and draughty rooms.
4. Carbon dioxide is known to _____ depletion of the ozone layer.
5. Heating water to 100 °C will _____ about the formation of air bubbles in the water.
6. New laws _____ rare species from becoming extinct.
7. In developing countries a lot of diseases are _____ by the lack of clean water.
8. Acid rain _____ from the use of fossil fuels.
9. Dumping pollutants into the ocean can _____ to a serious ecological problem.
10. Fitting cars with catalysts _____ about significant reductions in carbon dioxide emissions.
11. Heat _____ water to evaporate.
12. Air pollution could _____ a number of different sources.
13. Good insulation _____ heat loss in the home.
14. The ozone layer _____ the earth from UV-radiation.
15. The increase in the world population _____ a global water crisis.

Exercise 2 Formulate sentences with the phrases below.

1. ozone hole / CFCs in the atmosphere
2. filters / floating debris / water supply
3. recycling household waste / landfill sites / to overflow
4. refrigerator / food / to spoil
5. good planning / production costs / down
6. dumping waste / ground and water pollution
7. proper insulation of an electric device / user / to get a shock
8. phosphates found in detergents / algae / to develop / rapid / in lakes
9. the widespread use of battery-operated vehicles / improvement in air quality ?

Other connectives linked to causal relationships

WHEN + clause

When warm water returns to the storage tank, it displaces the colder water at the bottom.

IF + clause (+ WILL) = predicting a result

If a house is insulated properly, heating costs are significantly reduced.

If you use energy efficiently you can save a lot of money and help to conserve natural resources.

It is easy to improve the standard of insulation **if** you are advised by a specialist.

If ice is heated to melting point, it begins to liquefy. (....., it **will** begin to liquefy.)

If the windows are replaced by double-glazed ones, a lot of heat is retained. (...., **will** be retained...)

BECAUSE
SINCE + clause
AS
AS THE/A RESULT OF

Since the government has introduced new legislation, the use of disposable packaging has been significantly reduced.

Solar hot water systems have become an ecological alternative to traditional heating systems **as a result of** constant technological development and improvement.

As sunlight heats the collector, heat is transferred to the water in the tank.

BECAUSE OF
ON ACCOUNT OF + noun
OWING TO
DUE TO

Due to recycling initiatives, the amount of waste has been reduced by almost 50%.

THEREFORE / CONSEQUENTLY / THUS / AS A RESULT / FOR THIS REASON / SO / THEREBY

The forces between the molecules of thermosetting polymers are very strong and **therefore** (**thus/consequently/hence**) they are difficult to recycle.

The forces between the molecules of thermosetting polymers are very strong. **As a result (for this reason/so)**, they are difficult to recycle.

Thermosetting polymers are difficult to recycle, **thereby** contributing to the ever increasing problem of waste management.

SO / SO THAT + RESULT

Thermoplastic polymers are heated up **so that** the material becomes soft and can be shaped into different forms.

Exercise 3¹

Complete the text on the most common causes and effects of machine vibration.

due to.....lead to.....could be caused..... may be caused.....
...produce... cause (3x) result

Machine Vibration

Example: Vibration can result from a number of conditions, acting alone or in combination.

Imbalance: A “heavy spot” in a rotating component will _____ vibration when the unbalanced weight rotates around the machine’s axis, creating a centrifugal force. Imbalance _____ by manufacturing defects like machining errors or casting flaws or maintenance issues such as deformed or dirty fan blades.

Misalignment: Vibration can _____ when machine shafts are out of line. Angular misalignment occurs when the axes of, for example a motor and pump are not parallel. Misalignment _____ during assembly or develop over time, _____ thermal expansion, components shifting, or improper reassembly after maintenance.

Wear: When a roller bearing race becomes pitted, for instance, the bearing rollers will _____ a vibration each time they travel over damaged area. A gear tooth that is heavily chipped or worn, or a drive belt that is breaking down, can also _____ vibration.

The effects of vibration can be severe. Unchecked machine vibration can accelerate rates of wear, i.e. reduce bearing life, and damage equipment. Vibrating machinery can create noise, _____ safety problems and _____ degradation in plant working conditions.

¹ Source reference: Adapted from: engine – Englisch für Ingenieure (2/2013, p.20) Title of the article: Machine Vibration, Auszug eine White Paper der Firma Fluke.